SECOND BRAIN LABORATORIES · AUTONOMOUS SYSTEMS

The Autonomous Legal Second Brain

Endogenous Knowledge Navigation

A synthetic Mexican legal research laboratory: from interconnected sources to governed judgment

A 16-cell pedagogical experiment integrating semantic retrieval, five knowledge graphs, autonomous navigation, evidence validation and committee memoranda.

Alejandro Reynoso · October 2026

The central proposition: retrieval is not judgment



Retrieval asks

Which sources resemble the legal question?

Autonomous research asks

What evidence is missing, which relationship should be explored, and when should research stop?

Relevance is not legal authority. An LLM recommendation is not a committee decision.

The complete laboratory: problem, method, outcome

Sixteen cells, one research architecture

Pedagogical explanation

This presentation follows a legal research question through a bounded system rather than treating retrieval as the whole task. A fixed synthetic vault supplies records and relationships; semantic search proposes starting evidence; and an agent may choose a next step from a controlled graph frontier. Deterministic checks validate identifiers, enforce budgets, and preserve an inspectable trace. The evaluation stages compare what different methods find, what they miss, and what resources they use. Memorandum drafting comes last, downstream of the evidence record. Separating these functions helps students locate errors and discuss where human review remains essential.

Research checkpoints

- **Problem:** legal questions may cross subject domains, document types, procedures, and time periods.
- **Method:** a synthetic corpus, typed graph, semantic index, bounded agent, and deterministic controls.
- **Outputs:** research traces, evidence inventories, comparative diagnostics, and draft committee memoranda.
- **Review:** distinguish intended design, actual execution, and verified findings.

Teaching takeaway. The system automates bounded investigation steps; it does not assume legal authority or transfer approval responsibility away from people.

Cell 2: construct the synthetic legal corpus

350 fictional records across ten domains

Pedagogical explanation

The 350 fictional records form the fixed evidence base for every method. Each record has a domain-coded ID, type, two controlled topics, a status flag, and short text. A canonical JSON file holds records constant across runs. Status values model illustrative, exception-sensitive, and superseded material without reproducing Mexican legislative history. The uneven domain allocation also lets students examine how corpus composition can affect similarity rankings and measured coverage.

Research checkpoints

- **Allocation:** constitutional 25, civil 40, commercial 45, criminal 35, and tax 45.
- **Allocation:** administrative 35, securities 35, customs 25, international 30, and jurisprudence 35.
- Two labels come from 20 topics, including financing, disclosure, due process, and share transfer.
- Save the canonical JSON; validate unique IDs, complete fields, 350 records, and synthetic markers.

Teaching takeaway. The corpus defines the experiment's evidence boundary. Fictional records must never be cited as enacted Mexican law.

Cell 3: interconnect the legal knowledge graph

Five typed relationship layers

Pedagogical explanation

The graph changes independent notes into a navigable research environment. Its multigraph structure preserves multiple relationship types between the same records. The five layers are hierarchy, substantive, interpretation, procedure, and cross-border. Deterministic rules create nearby intra-domain links, constitutional and jurisprudential connections, thematic bridges, and selected cross-domain links. The resulting graph and manifest support audit, while Markdown notes expose typed relationships as Obsidian links. These edges are navigational cues, not proof of legal hierarchy or applicability.

Research checkpoints

- **Hierarchy and substance:** model possible dependencies and shared legal subjects.
- **Interpretation and procedure:** connect jurisprudence and institutional process.
- **Cross-border:** expose international or regulatory relationships.
- **Outputs:** NetworkX graph, edge manifest, corpus JSON, and linked Markdown notes.

Teaching takeaway. A graph makes relationships visible; it does not create legal authority, normative precedence, or applicability.

Cell 4: build semantic and graph-aware retrieval

Relevance proposes candidates for review

Pedagogical explanation

Retrieval is shared by the baselines and agent. TF–IDF represents document text and topics using unigrams and bigrams, English stop-word removal, and a 6,000-feature ceiling. Cosine similarity ranks records against the consultation; direct inspection returns a selected record's text and metadata. A separate graph-frontier tool lists unvisited neighbors. Keeping ranking distinct from navigation helps diagnose failures: an omitted result may reflect indexing or query wording, while a visible but unvisited neighbor may reflect path selection. Scores indicate resemblance under chosen features, never legal authority or sufficiency.

Research checkpoints

- **Index:** sparse vectors encode 350 records with up to 6,000 terms.
- **Search:** return top-ranked candidates by cosine similarity.
- **Frontier:** expose unvisited neighbors with IDs, domains, scores, and relation layers.
- **Review:** ranking is a candidate aid; inspect text and metadata directly.

Teaching takeaway. A similarity score is a search signal, not a legal conclusion. Language, metadata, and corpus composition can shape the ranking.

Cell 5: define three legal consultations

Multidisciplinary scenarios with hidden evaluation keys

Pedagogical explanation

The consultations test whether a method can move beyond one keyword or domain. Case A concerns foreign-investor financing, disclosure, and contractual breach. Case B concerns disputed import classification, a customs declaration, and a tax assessment. Case C concerns a cross-border acquisition, share transfer, financing, and regulatory filings. Each case spans five target domains. The highest-similarity synthetic record in each target domain can serve as a hidden instructor key. These labels support evaluation; they are not legal ground truth.

Research checkpoints

- **Case A:** commercial, securities, tax, administrative, and constitutional domains.
- **Case B:** customs, tax, administrative, criminal, and constitutional domains.
- **Case C:** commercial, international, securities, tax, and civil domains.
- Select one high-similarity key per target domain; withhold keys from the agent.

Teaching takeaway. Target domains measure designed coverage. Instructor keys support comparison but do not establish legal correctness or exclude alternative evidence.

Cell 6: establish two non-agent baselines

Separate adaptive judgment from simpler methods

Pedagogical explanation

The baselines show what can be achieved without a language model choosing the next research action. Semantic retrieval returns the highest-ranked records directly. Fixed traversal starts from the top three semantic seeds and explores sorted graph neighbors breadth-first, subject to a depth limit and source budget. Both methods use the same corpus and consultation as the autonomous method. Their paths and visited sets support comparisons of coverage and sequence. Equal source budgets improve fairness but do not equalize API calls, latency, or computational cost; report those resources separately.

Research checkpoints

- **Semantic search:** select top-k records by TF–IDF cosine similarity only.
- **Graph expansion:** traverse breadth-first from three seeds to a fixed depth.
- **Comparable limits:** apply a shared maximum inspected-document budget.
- **Outputs:** ranked list or deterministic visited set, with paths and counts.

Teaching takeaway. A useful baseline is a fair, reproducible alternative—not a deliberately weak method. Equal source budgets do not imply equal total cost.

Cell 7: select the initial research seeds

A starting point, not a predetermined path

Pedagogical explanation

The seed stage initializes each case with the three highest-ranked semantic records. It preserves the question, visited IDs, ordered path, trace, unresolved issues, errors, and model-call count so later decisions have an explicit state. Seeds set the agent's initial field of view: a domain-specific phrase may produce different records from a paraphrase, and a narrow starting set may bias the frontier. Separating seeds from later visits lets evaluators test whether navigation broadens or compounds the initial perspective.

Research checkpoints

- Rank the corpus against the question and select the three highest-scoring records.
- Record each seed's ID, domain, similarity score, and role as an initial source.
- Treat domain diversity and topic coverage as review questions, not assumed outcomes.
- Compare seed domains with target domains before assessing later navigation.

Teaching takeaway. Cell 7 chooses where to begin. Cell 8 is evaluated on what it does with that starting state.

Cell 8: introduce autonomous research judgment

The agent selects from a constrained frontier

Pedagogical explanation

At each iteration, the agent receives the consultation, recent inspection metadata, open issues, candidate frontier, and remaining budget. Candidates include IDs, domains, similarity scores, and graph layers. The model must return a structured decision to visit one offered ID or stop, with a short reason. A deterministic executor validates the action, retrieves the record, updates state, and logs the result. Minimum-evidence rules, document and model-call limits, and explicit error handling bound the loop. An unavailable or invalid model response triggers a labeled heuristic, not a live-model result.

Research checkpoints

- **Payload:** question, recent visits, open issues, frontier candidates, and remaining limits.
- **Output:** JSON visit-or-stop action, offered node ID, and concise research reason.
- **Safeguards:** validate JSON and frontier membership; enforce evidence and budget rules.
- Log errors, tool results, stop reasons, and live versus heuristic execution mode.

Teaching takeaway. A reported rationale is inspectable evidence of system output, not proof of sound legal reasoning. Verify the live model path in Colab.

Cell 9: prepare the evidence memorandum

Make sources, status, and gaps reviewable

Pedagogical explanation

The evidence memorandum turns the agent's path into a structured research record. It combines the case, run mode, visited documents, domain coverage, source metadata, applicability flags, and questions needing human verification. Sections distinguish the consultation, method, domains investigated, selected evidence, applicability cautions, missing facts, and scope disclaimer. The citation audit checks that cited IDs belong to the visited set, improving provenance without establishing that a cited source supports a claim. This is an inventory for review, not a final legal opinion.

Research checkpoints

- **Inputs:** case facts and targets, agent path and stop reason, document metadata.
- **Evidence:** visited IDs, domains, key topics, and synthetic status flags.
- **Cautions:** identify exception-sensitive or superseded records for follow-up.
- **Human review:** confirm facts, dates, authority, procedure, and official sources.

Teaching takeaway. Citation presence is not citation support. Provenance checks help reviewers inspect the record but cannot validate the legal analysis.

Cells 10–14: evaluate the research process

Five complementary tests, not one success score

Pedagogical explanation

Evaluation asks whether adaptive navigation improves the evidence gathered, how that result depends on resources, and where the system remains fragile. These cells compare methods and domain coverage, inspect recovery of instructor-selected records, audit sufficiency and stopping, vary document budgets, and perturb consultation wording while reviewing traces. Metrics should be calculated from saved runs and reported with denominators and execution mode. A diagram or example number is not a measured result. Live-model and fallback outcomes represent different policies and should not be pooled without explicit labeling.

Research checkpoints

- **Comparison:** report visits, domain breadth, key recovery, topical range, and model calls.
- **Provenance:** examine source type, legal domain, status flags, and authority cautions.
- **Sufficiency:** compare the agent's stop rationale with explicit external criteria.
- **Sensitivity and robustness:** vary budgets and wording; inspect paths and open issues.

Teaching takeaway. No single metric certifies legal quality. Report observed results only after reproducible execution and keep illustrative designs distinct from findings.

Cell 15: compile the research dossier

Preserve the record needed for reconstruction

Pedagogical explanation

The dossier makes a run portable without stripping away how it was produced. A case folder can assemble an executive summary, full memorandum, evidence memorandum, visited-source table, graph subgraph, decision trace, key findings, unresolved issues, and source notes. Corpus version, configuration, model or fallback mode, budget, case, and run IDs preserve the context needed to reconstruct the path. Exports depend on the executed notebook version and available dependencies. Verify each expected file exists, opens, and matches the source records; do not infer successful output from a design diagram.

Research checkpoints

- **Research record:** consultation, inspected sources, citations, and applicability flags.
- **Analysis:** executive summary, developed memorandum, key findings, and open issues.
- **Traceability:** visited-source CSV, navigation JSON, graph summary, and corpus manifest.
- **Release:** organize by case, retain synthetic disclaimers, and verify every export.

Teaching takeaway. A complete dossier supports audit and reproducibility; professional review remains responsible for evaluating any real legal conclusion.

Cell 16: develop committee-style legal memoranda



Three draft memoranda: financing, customs and acquisition. Each is approximately 400–600 words and organizes facts, legal questions, fictional authorities, integrated analysis, uncertainties and a proposed committee resolution.

Writing is downstream of research

- Use only inspected synthetic evidence.
- Verify citations against visited IDs.
- Separate supported claims from unresolved questions.
- Preserve professional review and approval.

Drafting and formatting controls do not certify legal correctness.

From the synthetic laboratory to Mexican legal practice

Institutional requirements

- Verify current official legal texts and applicable temporal versions.
- Distinguish constitutional norms, statutes, regulations and jurisprudence.
- Confirm competence, jurisdiction and procedural posture.
- Record provenance, amendments, exceptions and legal effect.

Governance requirements

- Protect privileged and confidential client information.
- Preserve citations, tool calls, research limits and audit trails.
- Escalate uncertain or consequential legal conclusions.
- Require responsible lawyers and committees to validate conclusions.

A synthetic legal research agent is a pedagogical prototype, not a deployable Mexican legal adviser.

TOPIC 791 · CONCLUSIONS

From retrieval to autonomous inquiry

The system does not merely retrieve what it knows.
It decides what evidence to investigate next.

Autonomy must remain bounded by source validity, transparent reasoning, research budgets and institutional authority.

Alejandro Reynoso · October 2026